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2025-2026
MECE - 21359763

Power Electronics For The Geec

Project Overview

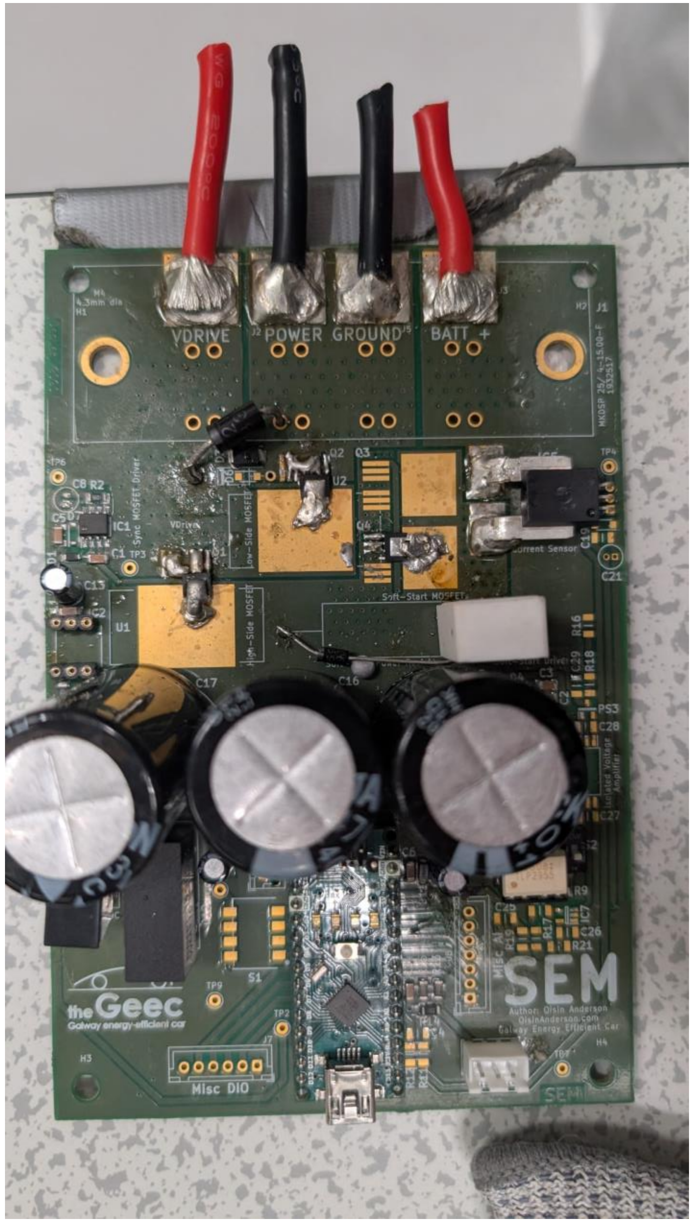
- The GEEC (Galway Energy Efficient Car) is a participant in the Shell Eco Marathon.
- It’s entered in the prototype lithium-ion class.
- The highest score GEEC achieved was 354km/kWh or 23c to drive from here to Dublin.

Power Electronics Requirements:

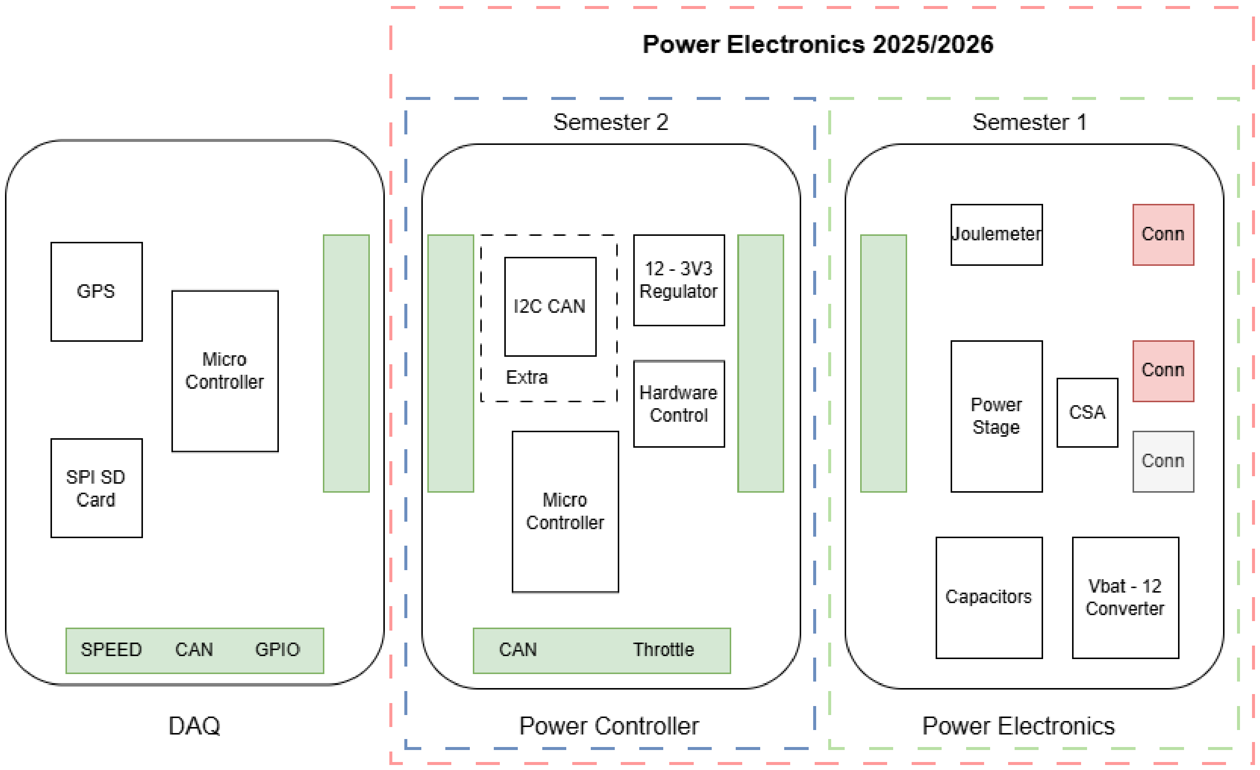
- Drive an Electric Motor.
- Take Throttle input from the driver
- **Fail Safe**

Current Power Electronics

- Inflexible
- Obsolete Components
- 2 Layer Design
- Large Loop Area, resulting in LC resonator
- Field Modifications



The Project Scope



Semester 1:

Power Electronics

- 4 layer SMD
- Full isolation

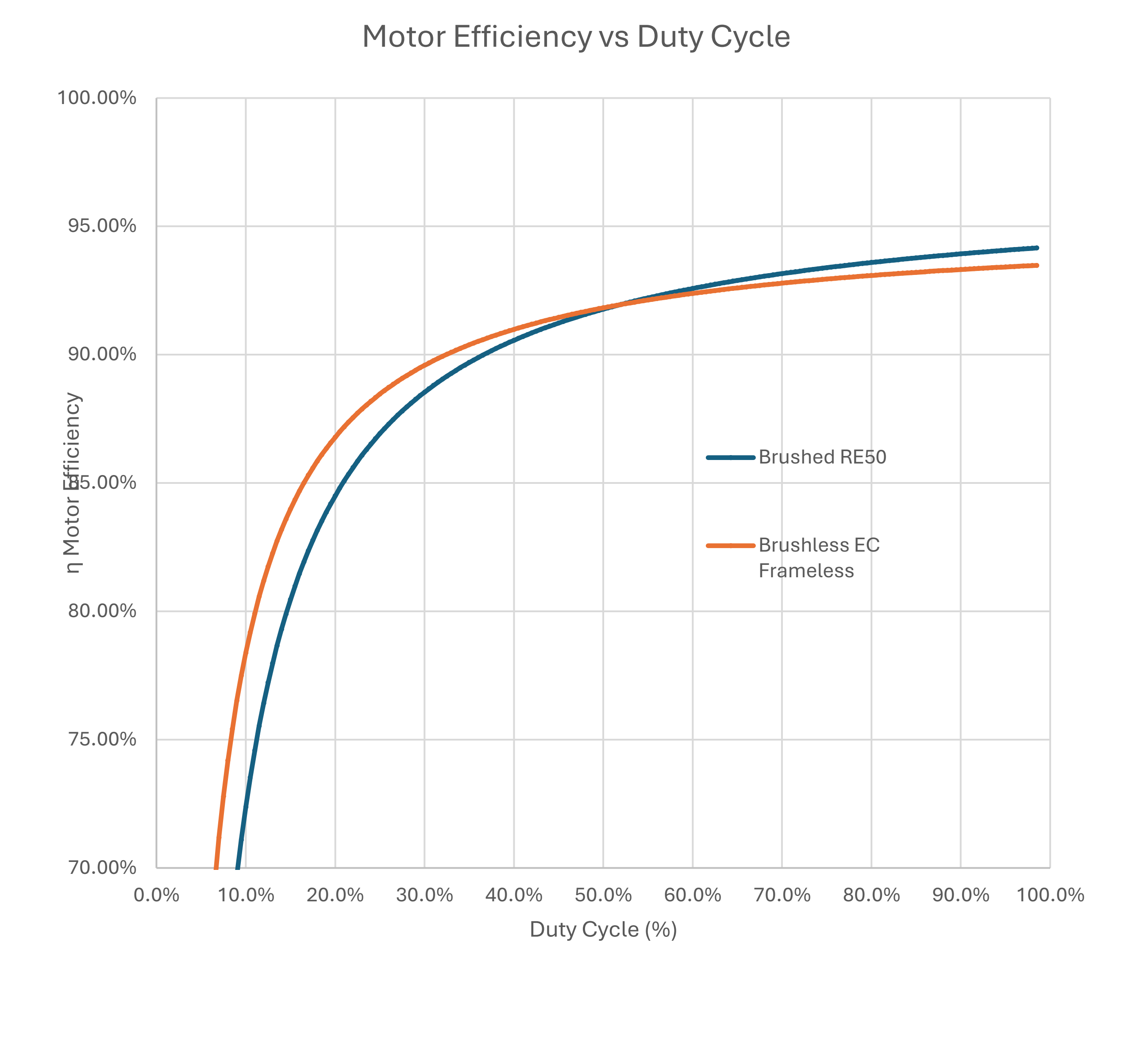
Semester 2:

Power Controller

- 2 layer THT
- Closed Loop Feedback

Brushless Motors

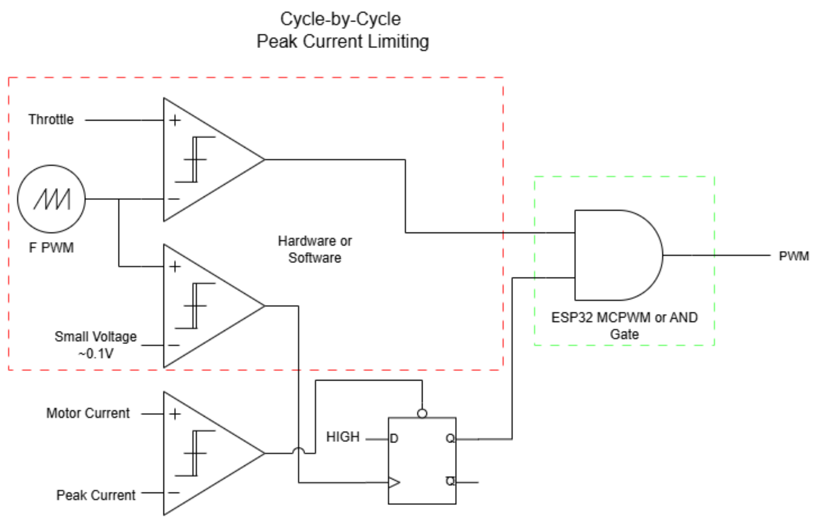
- Previous Projects Investigated Brushless Controllers & 48V control
- Theoretical Calculations show worse performance ~2%

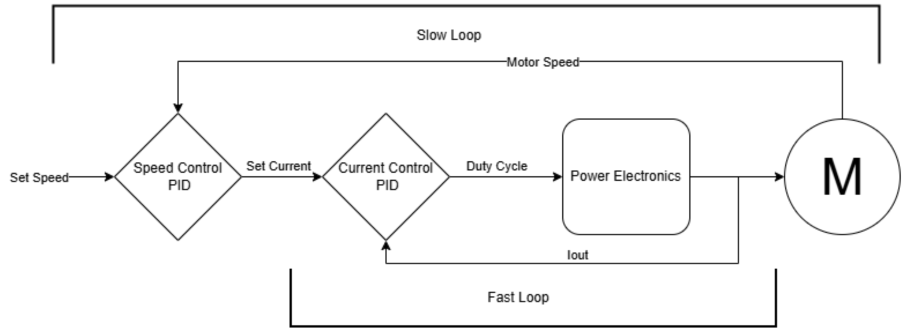


Current Control

- Hardware Limiting:
  - Cycle-by-Cycle
  - Latching

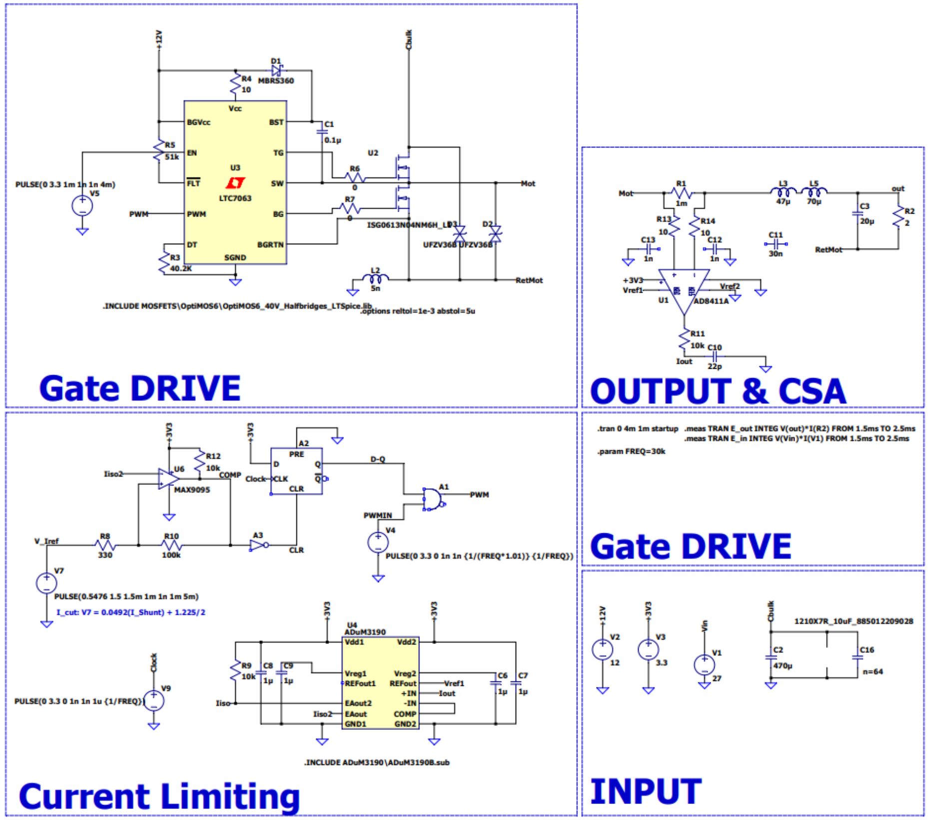
- Software Limiting
  - ~10Hz

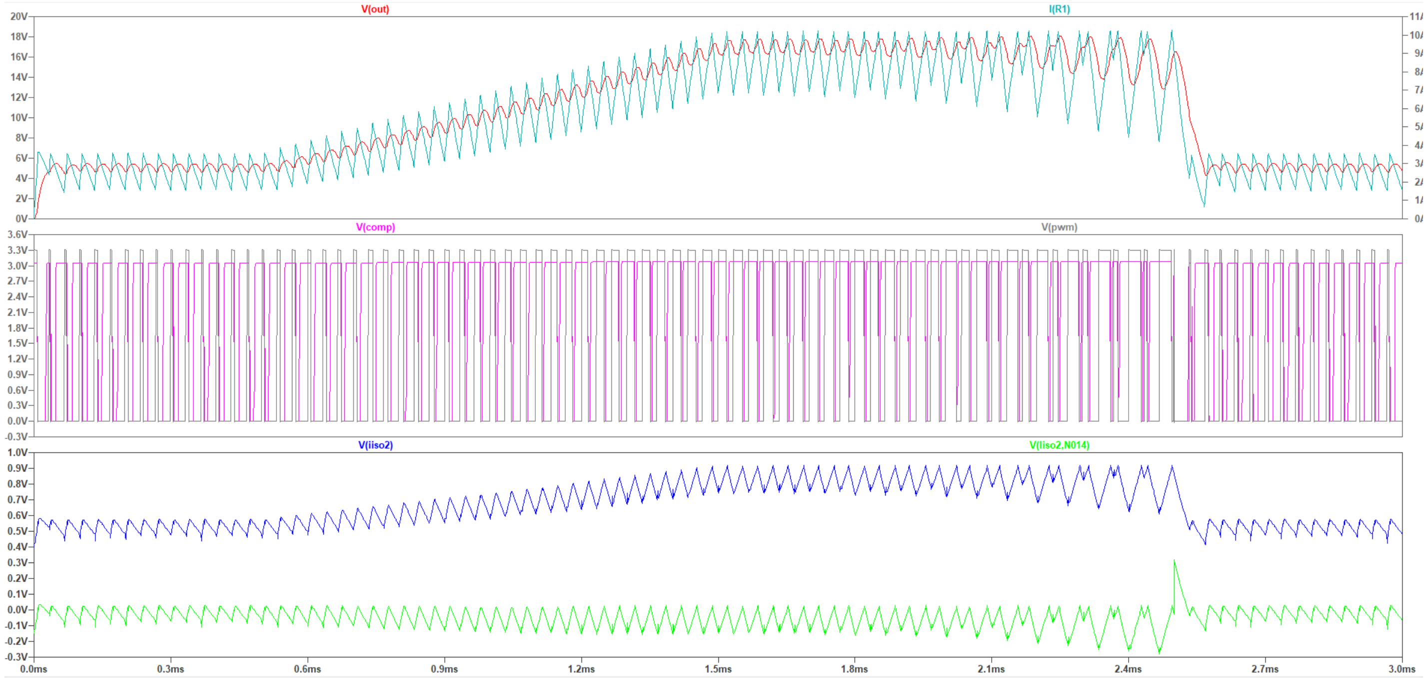




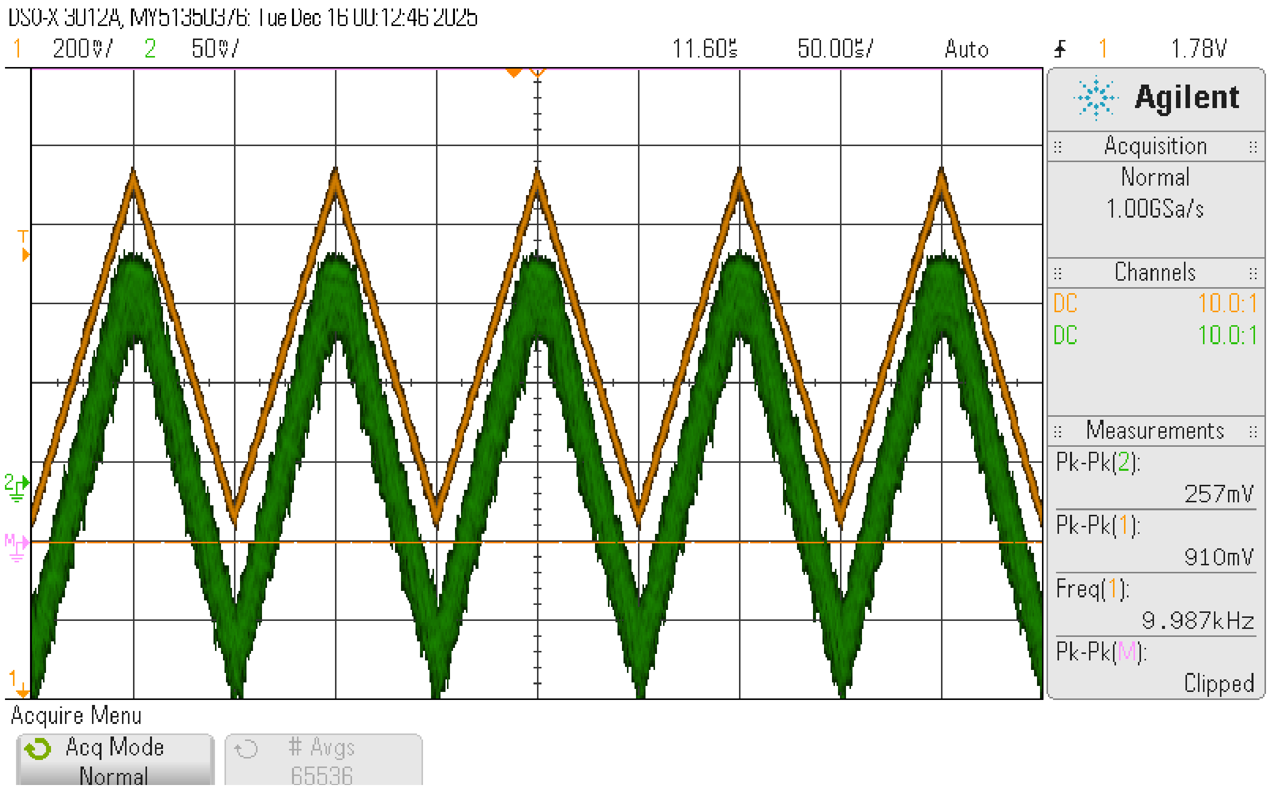
Simulation

- Closed Loop Current Control
- Variable Current Limit





Testing



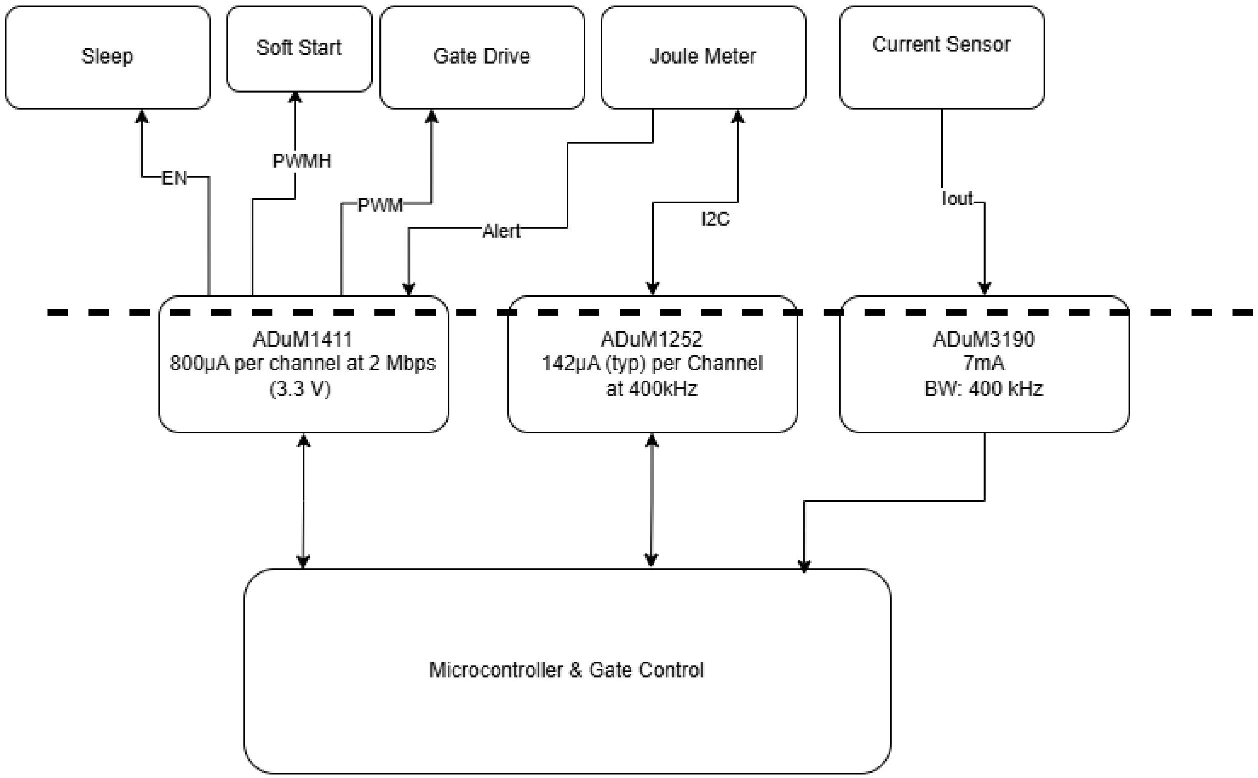
- Orange: AD8411A
- Green: ADuM3190

Expected A_V: 1
Actual A_V: 0.33

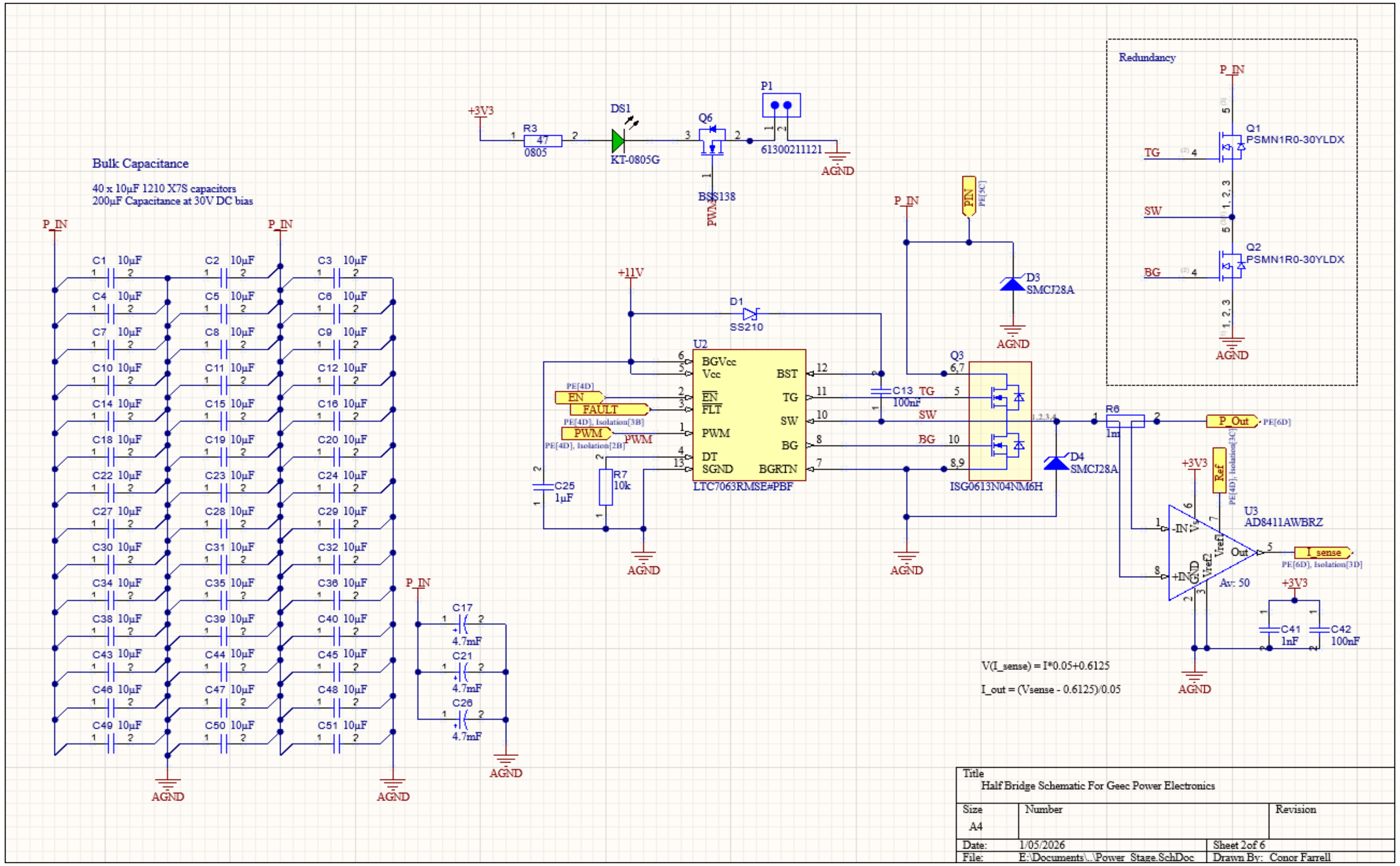
Solution: Op-Amp (A_V 2) and 800kHz low pass filter on Power Controller.

Protection

- Low power digital and analogue isolators.
  - Consumes 46mW
- TVS Diodes to shunt 15A at 38V



Schematic



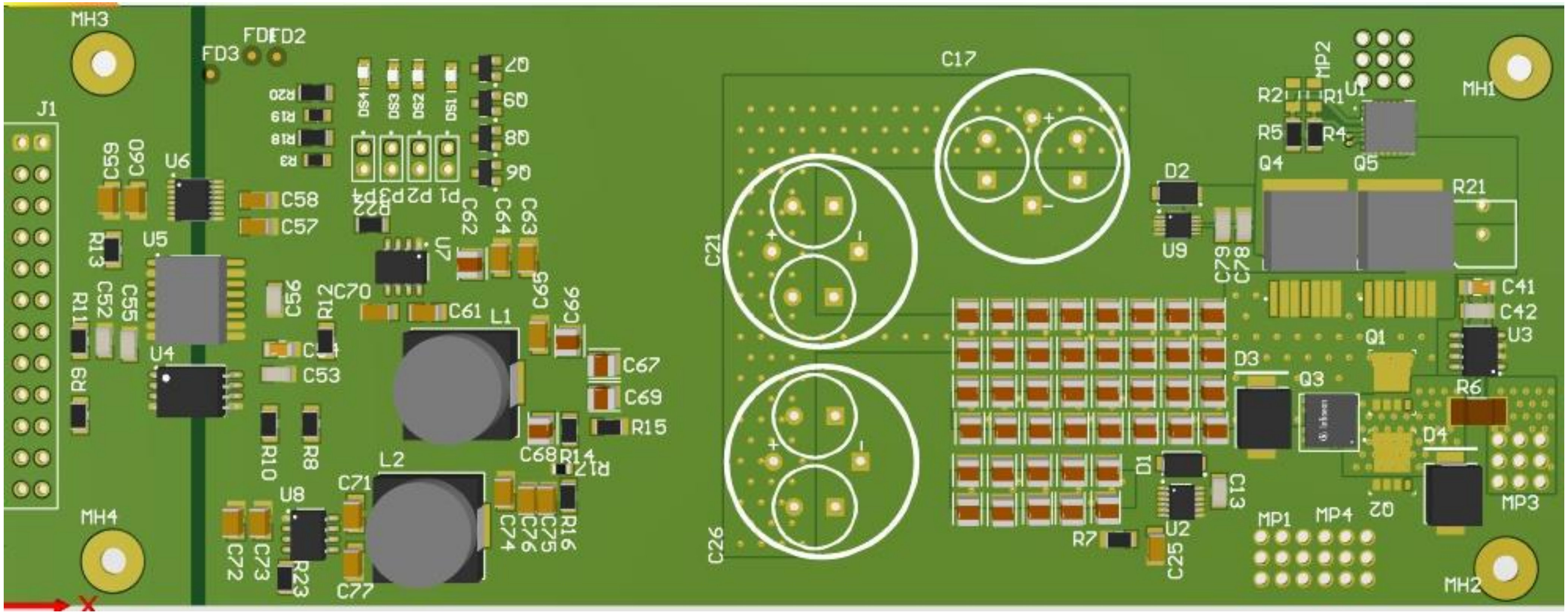
Half Bridge Design

- 40 10µF MLCC Capacitors & 3 4.7mF Electrolytic Capacitors
- Redundant MOSFETS for easy Replacement
- TVS Diodes Capable of shunting 15A @ 38V
- Indicator LED with removable jumper.

PCB Design

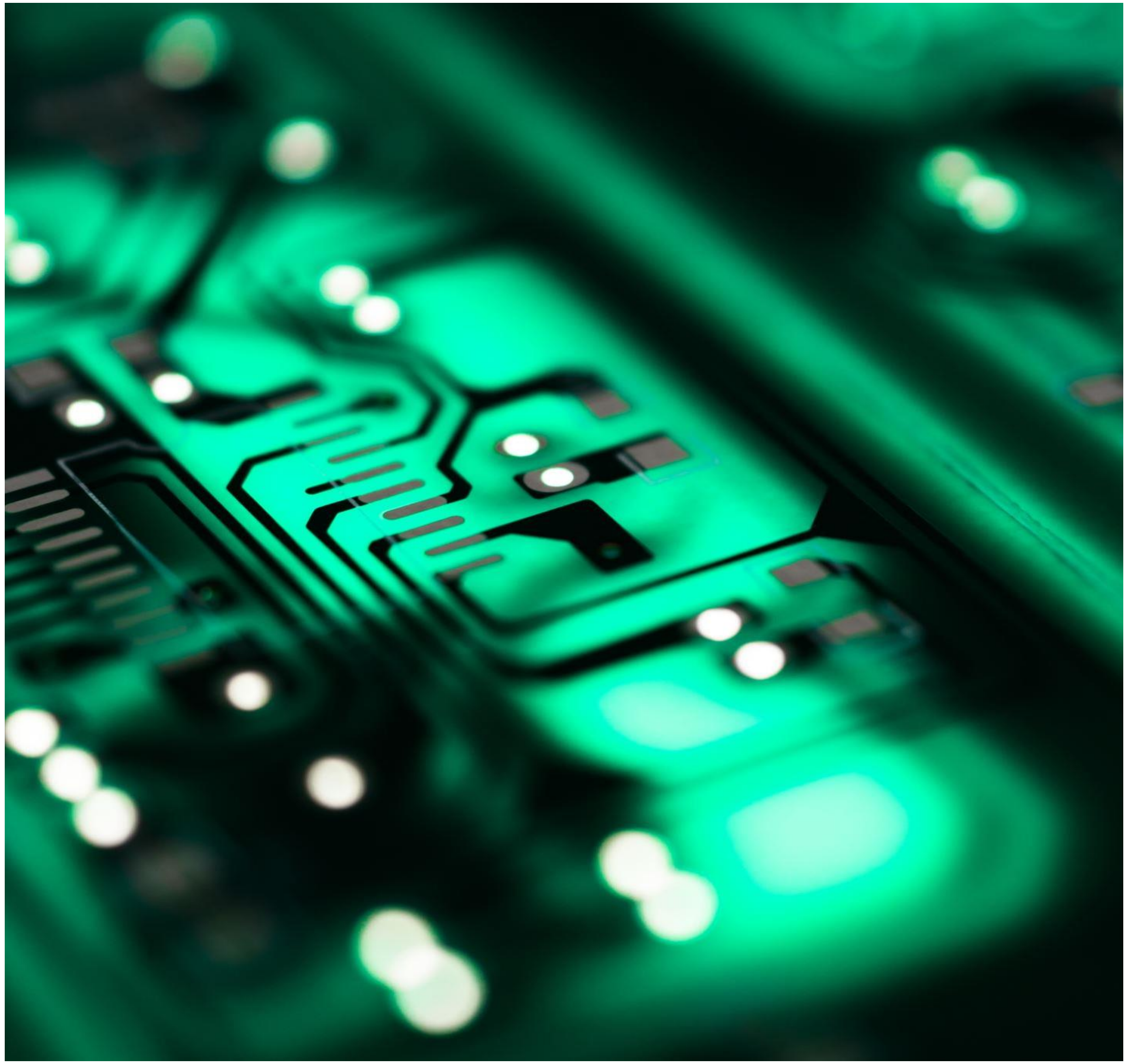
- 4 Layer Board, Isolated Ground planes
- Via Stitching on High current planes
- Minimisation of loop area and change in loop area

Preliminary Design



This Semester

- Design Power Controller Board
- Assist in Design of DAQ Board
- Testing of Hardware
- Integrate the Design into the DAQ
- Determine Performance Improvement



Thanks For Listening

Any Questions?